Arista Access Node BTM-ACC-19 Optical Transport Failure Standard Operating Procedure

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1298

Description

This Standard Operating Procedure (SOP) addresses a major optical transport failure on Arista Access Node BTM-ACC-19, located at BTM Layout Aggregation. The failure is characterized by Alarm Code ALARM_1298, indicating SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This failure domain occurs on the Access role due to the inherent vulnerability of optical interfaces to environmental stressors and aging effects. As optical signals traverse the fiber, they are susceptible to attenuation, dispersion, and nonlinear effects, leading to signal degradation. Furthermore, physical stressors such as temperature fluctuations, vibration, and mechanical shock can cause SFP/XFP degradation. To mitigate these risks, regular maintenance and inspection are essential. In the event of a failure, prompt action is required to minimize downtime and ensure service continuity. This SOP provides a step-by-step guide to diagnose and resolve the issue, ensuring a swift return to normal operations.

Pre-requisite Checks

- check: Verify the site location and node ID (BTM-ACC-19) to ensure the correct equipment is targeted., status: Complete

- check: Confirm the Access role and Arista vendor equipment to ensure the correct CLI dialect is used., status: Complete

- check: Inspect the physical environment for any signs of damage, moisture, or extreme temperatures., status: Complete

- check: Verify the redundant link status to ensure alternate paths are available., status: Complete

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to physical stressors or aging effects.	High
Fiber cuts or breaks due to mechanical damage or environmental stressors.	Medium
Rx/Tx power loss due to optical signal degradation or fiber attenuation.	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface | i ALARM_1298, Expected Output: ALARM_1298: SFP/XFP degradation or fiber cuts detected, Description: This command isolates the specific interface experiencing the optical transport failure, indicating whether SFP/XFP degradation or fiber cuts are the root cause.
```

```
Command: show interface | i Rx/Tx power loss, Expected Output: Rx/Tx power loss detected, Description: This command verifies the presence of Rx/Tx power loss, which may indicate optical signal degradation or fiber attenuation.

Command: show system internal optics | i , Expected Output: Optics status: Degraded or Faulty, Description: This command provides detailed optics information, including the current status and any error messages, to aid in diagnosing the issue.
```

Resolution Steps

Step 1: Step: Verify and replace the SFP/XFP module., Command: replace module with , Description: Replace the degraded or faulty SFP/XFP module with a new one to restore optical signal integrity.

Step 2: Step: Inspect and clean the fiber optic connections., Command: no shutdown, Description: Inspect the fiber optic connections for any signs of damage or contamination and clean them as necessary to ensure optimal signal transmission.

Step 3: Step: Verify and adjust the Tx/Rx power levels., Command: set interface tx-power , Description: Verify and adjust the Tx/Rx power levels to ensure optimal signal transmission and reception.

Post-Check Verification

```
Command: show interface , Expected Output: ALARM_1298: Cleared or Inactive, Description: Verify that the optical transport failure has been resolved and the alarm is cleared or inactive.

Command: show system internal optics | i , Expected Output: Optics status: Normal or Healthy, Description: Verify the optics status to ensure it is normal or healthy after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module., Command: replace module with , Description: Restore the original SFP/XFP module in case the new replacement module is faulty or incompatible.

- Step: Revert the Tx/Rx power level adjustments., Command: set interface tx-power , Description: Revert the Tx/Rx power level adjustments to their original values if the new settings are not optimal.